

Lecture 9: Temperature and Heat – Tutorial Sheet

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Use the Master Procedure: identify, sketch, convert units, predict, select the model, execute, evaluate. Use kelvin whenever T is raised to a power. Problems A1 and A2 are solved in full; final answers to the others are at the end. Useful values: $c_{\text{water}} = 4186$, $c_{\text{Cu}} = 390$, $c_{\text{Al}} = 900 \text{ J/(kg K)}$; $\alpha_{\text{steel}} = 1.2 \times 10^{-5}$, $\alpha_{\text{Al}} = 2.4 \times 10^{-5} \text{ K}^{-1}$; $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$.

A. Worked examples

A1. A metal ring on a rod

A steel rod of diameter 2.500 cm at 20°C must pass through an aluminium ring whose hole has diameter 2.490 cm at 20°C. To what temperature must the ring be heated (the rod stays at 20°C)?

Solution. The hole expands like a solid of aluminium: $d = d_0(1 + \alpha\Delta T)$. We need $2.500 = 2.490(1 + 2.4 \times 10^{-5} \Delta T)$, so $\Delta T = \frac{2.500/2.490 - 1}{2.4 \times 10^{-5}} = \frac{4.016 \times 10^{-3}}{2.4 \times 10^{-5}} = 167 \text{ K}$. The ring must reach $T = 20 + 167 \approx 190^\circ\text{C}$. *Check:* the hole grows by 0.010 cm, which is 0.40% of its diameter, about $167 \times 2.4 \times 10^{-5}$.

A2. Cooling a drink

A 0.250 kg glass of water at 25°C is cooled by dropping in an 0.040 kg aluminium block at -10°C (no phase change, ignore the glass). Find the final temperature.

Solution. $m_w c_w (T_f - 25) + m_{\text{Al}} c_{\text{Al}} (T_f + 10) = 0$: $(0.250)(4186)(T_f - 25) + (0.040)(900)(T_f + 10) = 0$, i.e. $1046.5(T_f - 25) + 36.0(T_f + 10) = 0$, so $T_f = \frac{26\,163 - 360}{1082.5} = 23.8^\circ\text{C}$. The result lies between -10 and 25°C and is close to 25°C because the water's heat capacity (1047 J/K) is much larger than the block's (36 J/K).

B. Practice

B1. Conversions

(a) Convert -40°C to $^\circ\text{F}$. (b) At what temperature do the Celsius and Fahrenheit scales agree? (c) Convert 20°C to kelvin and a change of 25°F to kelvin.

B2. A bridge

A steel bridge girder is 100.0 m long. What is its change in length between a winter night at -5°C and a summer afternoon at 40°C ?

B3. Thermal stress

A steel rod ($Y = 2.0 \times 10^{11} \text{ Pa}$) is held between two rigid walls at 20°C . Find the stress when it is heated to 70°C .

B4. Energy to heat

How much energy is needed to heat 2.0 kg of aluminium from 20°C to 120°C? How long does a 500 W heater take?

B5. Conduction

A glass window of area 1.5 m² and thickness 4.0 mm ($k = 0.80 \text{ W}/(\text{m K})$) separates a room at 22°C from outside at 2°C. Find the heat current. How much energy is lost in an hour?

B6. Radiation

A hot plate of area 0.040 m² ($e = 0.90$) is at 400°C in a room at 20°C. Find the power it radiates and the net power it loses.

B7. Challenge: the method is not given

A 0.50 kg iron pan ($c = 450 \text{ J}/(\text{kg K})$) at 200°C is dropped into 2.0 kg of water at 20°C. (a) Find the final temperature (assume no boiling). (b) The container, with walls of area 0.10 m² and thickness 2.0 cm ($k = 0.040 \text{ W}/(\text{m K})$), then loses energy to 20°C surroundings. Estimate the initial heat current and how long the water takes to cool by 1.0 K.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. A steel rod of length 2.50 m at 10 °C is heated to 45 °C ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$). Its increase in length is

- (A) 0.11 mm (B) 1.05 mm (C) 1.35 mm (D) 3.15 mm (E) 10.50 mm

E2. A block of aluminium has volume 0.0200 m³ (linear expansion coefficient $\alpha = 2.4 \times 10^{-5} \text{ K}^{-1}$). When it is heated by 60 K its volume increases by

- (A) 8.6 cm³
(B) 28.8 cm³
(C) 86.4 cm³
(D) 259.2 cm³
(E) 864.0 cm³

E3. A 500 W immersion heater delivers all its energy to 0.80 kg of water ($c = 4190 \text{ J kg}^{-1} \text{ K}^{-1}$),

raising its temperature from 20°C to 70°C . The heating time is

- (A) 5.59 s
- (B) 168 s
- (C) 335 s
- (D) 670 s
- (E) 1.68×10^5 s

E4. During one afternoon the temperature of a room rises by 36°F . This rise, expressed in kelvin, is

- (A) 2.2 K (B) 20.0 K (C) 36.0 K (D) 96.8 K (E) 293.1 K

Items E5 and E6 share this information: A glass window pane of area 2.0 m^2 and thickness 1.0 cm has a temperature difference of 15 K between its faces. The thermal conductivity of the glass is $0.80\text{ W m}^{-1}\text{K}^{-1}$. Steady state.

E5. The rate of heat conduction through the pane is

- (A) 24.0 W
- (B) 240 W
- (C) 4800 W
- (D) 2.40×10^5 W
- (E) 2400 W

E6. The energy conducted through the pane in 10 minutes is

- (A) 4.00 J
- (B) 2.40×10^4 J
- (C) 1.44×10^5 J
- (D) 1.44×10^6 J
- (E) 2.07×10^8 J

E7. A sphere of radius 0.10 m , emissivity 0.90 and surface temperature 800 K is in surroundings at 300 K . The net rate at which it radiates energy is ($\sigma = 5.67 \times 10^{-8}\text{ W m}^{-2}\text{K}^{-4}$)

- (A) 401 W (B) 494 W (C) 644 W (D) 2575 W (E) 2627 W

E8. Which of the following statements is/are true?

- I. A temperature change of 1°C is equal to a temperature change of 1 K .
- II. A hot body contains a large amount of heat Q , which is a property of its state.
- III. Two bodies in thermal equilibrium with each other are at the same temperature.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E9. For steady conduction, $H = -kA \, dT/dx$. Which of the following statements about the minus sign is/are true?

- I. It shows that heat flows in the direction of decreasing temperature.
- II. If T decreases as x increases, then $dT/dx < 0$ and $H > 0$ in the $+x$ direction.
- III. It shows that the thermal conductivity k is negative for good insulators.

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E10. A bottle wrapped in a wet cloth stays cooler than an identical dry bottle in a breeze. The best explanation is that

- (A) The faster molecules leave the liquid, so the average kinetic energy of those that remain falls and the liquid cools.
- (B) Evaporation adds heat to the liquid, which is why the surroundings become warmer.
- (C) The molecules that leave are the slowest, so the average speed of those that remain increases.
- (D) The water gives energy to the air only by radiation, and evaporation plays no part.
- (E) The temperature of a liquid cannot change while it evaporates.

Answers

- B1 (a) -40°F ; (b) -40 ; (c) 293.15 K ; $\Delta T = 14 \text{ K}$
 B2 $\Delta L = (1.2 \times 10^{-5})(100.0)(45) = 5.4 \times 10^{-2} \text{ m} = 54 \text{ mm}$
 B3 $Y\alpha\Delta T = 1.2 \times 10^8 \text{ Pa}$
 B4 $Q = 1.8 \times 10^5 \text{ J}$; $t = 360 \text{ s}$ (6.0 min)
 B5 $H = kA\Delta T/L = 6.0 \times 10^3 \text{ W}$; $2.2 \times 10^7 \text{ J per hour}$
 B6 emitted $H = e\sigma AT^4 = 4.2 \times 10^2 \text{ W}$ ($T = 673 \text{ K}$); net $4.0 \times 10^2 \text{ W}$
 B7 (a) $T_f = 24.7^\circ\text{C}$; (b) $\Delta T = 4.7 \text{ K}$, $H \approx 0.94 \text{ W}$; $t \approx 8.9 \times 10^3 \text{ s}$ (2.5 h)

Exam-style practice: answer key and brief solutions

Key: E1 B E2 C E3 C E4 B E5 E E6 D E7 D E8 C E9 B E10 A

E1 (B). $\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(2.50)(35) = 1.05 \times 10^{-3} \text{ m} = 1.05 \text{ mm}$. Use the temperature change $\Delta T = 35 \text{ K}$, not the final temperature (which would give 1.35 mm).

E2 (C). $\beta = 3\alpha = 7.2 \times 10^{-5} \text{ K}^{-1}$, so $\Delta V = \beta V_0 \Delta T = (7.2 \times 10^{-5})(0.0200)(60) = 8.64 \times 10^{-5} \text{ m}^3 = 86.4 \text{ cm}^3$. The linear coefficient α alone would give 28.8 cm^3 ; and $1 \text{ m}^3 = 10^6 \text{ cm}^3$.

E3 (C). $Q = mc\Delta T = (0.80)(4190)(50) = 1.68 \times 10^5 \text{ J}$ and $t = Q/P = (1.68 \times 10^5)/500 = 335 \text{ s}$, about 5.6 minutes. The value 5.59 is the time in minutes, and 1.68×10^5 is the energy.

E4 (B). A temperature *difference* converts by the ratio only, with no offset: $\Delta T_K = \Delta T_C = \frac{5}{9}\Delta T_F = \frac{5}{9}(36) = 20 \text{ K}$. The offsets (32 and 273.15) apply to temperature *readings*, not to differences.

E5 (E). $H = kA \frac{\Delta T}{L} = (0.80)(2.0) \frac{15}{0.010} = 2400 \text{ W}$. Convert L to metres ($1.0 \text{ cm} = 0.010 \text{ m}$). Omitting $1/L$ gives 24 W ; using $L = 0.10$ gives 240 W .

E6 (D). Convert the time: $10 \text{ min} = 600 \text{ s}$. $Q = Ht = (2400)(600) = 1.44 \times 10^6 \text{ J}$, using the previous result. Using $t = 10$ gives only $2.4 \times 10^4 \text{ J}$.

E7 (D). $P_{\text{net}} = e\sigma A(T^4 - T_0^4)$ with $A = 4\pi r^2 = 0.126 \text{ m}^2$ and temperatures in kelvin: $= (0.90)(5.67 \times 10^{-8})(0.126)(4.096 \times 10^{11} - 8.1 \times 10^9) = 2575 \text{ W}$. The emitted power alone is 2627 W ; the surroundings return 52 W .

E8 (C). I is true (the degree sizes are equal). II is false: heat is energy *in transit* because of a temperature difference; a body contains internal energy, not heat. III is true (zeroth law). Answer: I and III only.

E9 (B). I is true: heat flows from hot to cold, so it is directed opposite to the temperature gradient. II is true: a negative gradient and the minus sign give a positive H along $+x$. III is false: $k > 0$ for every material (it is large for conductors and small for insulators). Answer: I and II only.

E10 (A). Evaporation removes the most energetic molecules from the liquid; the latent heat needed to evaporate comes from the liquid and its surroundings, so the remaining liquid (and the bottle) is cooled. B has the energy flow backwards, C reverses which molecules escape, D denies evaporation, and E confuses evaporation (any temperature) with boiling at constant temperature.